

Operatorul Bernstein

Notatie $e_k(x) = x^k$

Fie $f : [0, 1] \rightarrow \mathbb{R}$, operatorul B_n definit prin

$$B_n(f)(x) = \sum_{k=0}^n p_{n,k}(x) f\left(\frac{k}{n}\right),$$

unde

$$p_{n,k}(x) = C_n^k x^k (1-x)^{n-k}$$

se numește *operatorul Bernstein*, iar polinomul $B_n(f)$ se numește *polinomul Bernstein*.

1. B_n este un operator liniar ($B_n(\alpha f + \beta g) = \alpha B_n(f) + \beta B_n(g)$)
2. B_n este un operator pozitiv ($f(x) \geq 0 \Rightarrow B_n(f)(x) \geq 0, \forall x \in [0, 1]$)
3. (a) $B_n(e_0)(x) = e_0(x) = 1$
(b) $B_n(e_1)(x) = e_1(x) = x$
(c) $B_n(e_2)(x) = e_2(x) = x^2 + \frac{x(1-x)}{n} = \frac{n-1}{n}e_2(x) + \frac{1}{n}e_0(x)$
(d) Dacă $f \in C[0, 1]$ atunci $B_n(f) \Rightarrow f$ (convergență uniformă) pe $[0, 1]$ când $n \rightarrow \infty$

Teorema Bohman-Popoviciu-Korovkin

Fie $L_m : C[a, b] \rightarrow C[a, b]$ un șir de operatori liniari și pozitivi care verifică

$$\begin{aligned} L_m(e_0)(x) &= 1 + u_m(x) \\ L_m(e_1)(x) &= x + v_m(x) \\ L_m(e_2)(x) &= x^2 + w_m(x) \end{aligned}$$

și $\lim_{m \rightarrow \infty} u_m(x) = \lim_{m \rightarrow \infty} v_m(x) = \lim_{m \rightarrow \infty} w_m(x) = 0$ uniform pe $[a, b]$. Atunci pentru orice $f \in C[a, b]$ avem

$$\lim_{m \rightarrow \infty} L_m(f)(x) = f(x)$$

uniform pe $[a, b]$.

1. **Să se arate că șirurile de operatori de mai jos sunt șiruri de aproximare**

(a) $(M_n)_{n \in \mathbb{N}}, M_n : C[0, 1] \rightarrow C[0, 1]$ definiți prin

$$M_n(f)(x) = (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t f(t) p_{n,k}(t) dt}{k+1}$$

unde $p_{n,k}(t)$ sunt polinoamele fundamentale Bernstein

(b) $(K_n)_{n \in \mathbb{N}}$, $K_n : C[0, 1] \rightarrow C[0, 1]$ **definiți prin**

$$K_n(f)(x) = (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} f(t) dt$$

(c) $(U_n)_{n \in \mathbb{N}}$, $U_n : C[0, 1] \rightarrow C[0, 1]$ **definiți prin**

$$U_n(f)(x) = \frac{1}{3}B_n(f)(x) + \frac{2}{3}(n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} f(t) dt$$

Rezolvare: Vom aplica teorema Bohman-Popoviciu-Korovkin

(a)

$$\begin{aligned} M_n(\alpha f + \beta g)(x) &= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t(\alpha f + \beta g)(t) p_{n,k}(t) dt}{k+1} = \\ &= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \left(\frac{\int_0^1 t\alpha f(t) p_{n,k}(t) dt}{k+1} + \frac{\int_0^1 t\beta g(t) p_{n,k}(t) dt}{k+1} \right) = \\ &= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \left(\frac{\alpha \int_0^1 t f(t) p_{n,k}(t) dt}{k+1} + \frac{\beta \int_0^1 t g(t) p_{n,k}(t) dt}{k+1} \right) = \\ &= \alpha(n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t f(t) p_{n,k}(t) dt}{k+1} + \\ &\quad + \beta(n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t g(t) p_{n,k}(t) dt}{k+1} = \\ &= \alpha M_n(f)(x) + \beta M_n(g)(x), \quad \forall f, g \in C[0, 1], \quad \alpha, \beta \in \mathbb{R}, \quad x \in [0, 1]. \end{aligned}$$

Fie $f \in C[0, 1]$, $f \geq 0$, i.e. $f(x) \geq 0$, $\forall x \in [0, 1]$. Avem $xf(x) \geq 0$, $\forall x \in [0, 1] \Rightarrow \int_0^1 tf(t) dt \geq 0$.
Ținând cont de faptul că

$$p_{n,k}(x) = C_n^k x^k (1-x)^{n-k} \geq 0, \quad \forall x \in [0, 1]$$

deducem

$$M_n(f)(x) = \underbrace{(n+1)(n+2)}_{\geq 0} \underbrace{\sum_{k=0}^n p_{n,k}(x)}_{\geq 0} \underbrace{\frac{\int_0^1 t f(t) p_{n,k}(t) dt}{k+1}}_{\geq 0}$$

În continuare vom calcula $M_n(e_0)$, $M_n(e_1)$, $M_n(e_2)$.

$$\begin{aligned}
M_n(e_0)(x) &= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t e_0(t) p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 C_n^k t^{k+1} (1-t)^{n-k} dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\frac{n!}{(n-k)!k!} B(k+2, n-k+1)}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{n!}{(n-k)!(k+1)!} \frac{\Gamma(k+2)\Gamma(n-k+1)}{\Gamma(n+3)} = \\
&= \sum_{k=0}^n p_{n,k}(x) \frac{(n+2)!}{(n-k)!(k+1)!} \frac{(k+1)!(n-k)!}{(n+2)!} = \sum_{k=0}^n p_{n,k}(x) = \\
&= \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} = (x+1-x)^n = 1 = e_0(x)
\end{aligned}$$

$$\begin{aligned}
M_n(e_1)(x) &= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t e_1(t) p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t^2 p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 C_n^k t^{k+2} (1-t)^{n-k} dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\frac{n!}{(n-k)!k!} B(k+3, n-k+1)}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{n!}{(n-k)!(k+1)!} \frac{\Gamma(k+3)\Gamma(n-k+1)}{\Gamma(n+4)} = \\
&= \sum_{k=0}^n p_{n,k}(x) \frac{(n+2)!}{(n-k)!(k+1)!} \frac{(k+2)!(n-k)!}{(n+3)!} = \sum_{k=0}^n p_{n,k}(x) \frac{k+2}{n+3} = \\
&= \frac{1}{n+3} \sum_{k=0}^n p_{n,k}(x) (k+2) = \frac{n}{n+3} \left(\sum_{k=0}^n p_{n,k}(x) \frac{k}{n} + \frac{2}{n} \sum_{k=0}^n p_{n,k}(x) \right) = \\
&= \frac{n}{n+3} (B_n e_1)(x) + \frac{2}{n+3} (B_n e_0)(x) = \frac{n}{n+3} \cdot x + \frac{2}{n+3} \cdot 1
\end{aligned}$$

$$\begin{aligned}
M_n(e_2)(x) &= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t e_2(t) p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t^3 p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 C_n^k t^{k+3} (1-t)^{n-k} dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\frac{n!}{(n-k)!k!} B(k+4, n-k+1)}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{n!}{(n-k)!(k+1)!} \frac{\Gamma(k+4)\Gamma(n-k+1)}{\Gamma(n+5)} = \\
&= \sum_{k=0}^n p_{n,k}(x) \frac{(n+2)!}{(n-k)!(k+1)!} \frac{(k+3)!(n-k)!}{(n+4)!} = \sum_{k=0}^n p_{n,k}(x) \frac{(k+3)(k+4)}{(n+3)(n+4)} = \\
&= \frac{1}{(n+3)(n+4)} \sum_{k=0}^n p_{n,k}(x) (k^2 + 5k + 6) = \frac{n^2}{(n+3)(n+4)} \sum_{k=0}^n p_{n,k}(x) \frac{k^2}{n^2} + \\
&+ \frac{5n}{(n+3)(n+4)} \sum_{k=0}^n p_{n,k}(x) \frac{k}{n} + \frac{6}{(n+3)(n+4)} \sum_{k=0}^n p_{n,k}(x) = \\
&= \frac{n^2}{(n+3)(n+4)} (B_n e_2)(x) + \frac{5n}{(n+3)(n+4)} (B_n e_1)(x) + \frac{6}{(n+3)(n+4)} (B_n e_0)(x) = \\
&= \frac{n^2}{(n+3)(n+4)} \left(x^2 + \frac{x(1-x)}{n} \right) + \frac{5n}{(n+3)(n+4)} x + \frac{6}{(n+3)(n+4)}
\end{aligned}$$

Obtınem

$$\begin{aligned}
M_n(e_0) &= e_0 \\
M_n(e_1) &= e_1 + \left(-\frac{3}{n+3} e_1 + \frac{2}{n+3} e_0 \right) \\
M_n(e_2) &= e_2 + \left(-\frac{8n+12}{(n+3)(n+4)} e_2 + \frac{6n}{(n+3)(n+4)} e_1 + \frac{6}{(n+3)(n+4)} e_0 \right)
\end{aligned}$$

(b)

$$\begin{aligned}
K_n(\alpha f + \beta g)(x) &= (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} (\alpha f + \beta g)(t) dt = \\
&= (n+1) \sum_{k=0}^n p_{n,k}(x) \left(\alpha \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} f(t) dt + \beta \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} g(t) dt \right) = \\
&= \alpha(n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} f(t) dt + \beta(n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} g(t) dt = \\
&= \alpha(K_n f)(x) + \beta(K_n g)(x)
\end{aligned}$$

Fie $f \in C[0, 1]$, $f \geq 0$, i.e. $f(x) \geq 0, \forall x \in [0, 1]$. Avem

$$K_n(f)(x) = (n+1) \underbrace{\sum_{k=0}^n p_{n,k}(x)}_{\geq 0} \underbrace{\int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} f(t) dt}_{\geq 0} \geq 0$$

În continuare vom calcula $K_n(e_0)$, $K_n(e_1)$, $K_n(e_2)$.

$$\begin{aligned}
K_n(e_0)(x) &= (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} e_0(t) dt = (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} 1 dt = (n+1) \sum_{k=0}^n p_{n,k}(x) \cdot t \Big|_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} = \\
&= (n+1) \sum_{k=0}^n p_{n,k}(x) \frac{1}{n+1} = \sum_{k=0}^n p_{n,k}(x) = (B_n e_0)(x) = e_0(x) = 1 \\
\\
K_n(e_1)(x) &= (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} e_1(t) dt = (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} t dt = (n+1) \sum_{k=0}^n p_{n,k}(x) \cdot \frac{t^2}{2} \Big|_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} = \\
&= (n+1) \sum_{k=0}^n p_{n,k}(x) \frac{1}{2} \left(\frac{(k+1)^2}{(n+1)^2} - \frac{k^2}{(n+1)^2} \right) = \frac{n+1}{2} \sum_{k=0}^n p_{n,k}(x) \frac{2k+1}{(n+1)^2} = \frac{n}{n+1} \sum_{k=0}^n p_{n,k}(x) \frac{k}{n} + \\
&+ \frac{1}{2(n+1)} \sum_{k=0}^n p_{n,k}(x) = \frac{n}{n+1} (B_n e_1)(x) + \frac{1}{2(n+1)} (B_n e_0)(x) = \frac{n}{n+1} x + \frac{1}{2(n+1)} \\
\\
K_n(e_2)(x) &= (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} e_2(t) dt = (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} t^2 dt = (n+1) \sum_{k=0}^n p_{n,k}(x) \cdot \frac{t^3}{3} \Big|_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} = \\
&= (n+1) \sum_{k=0}^n p_{n,k}(x) \frac{1}{3} \left(\frac{(k+1)^3}{(n+1)^3} - \frac{k^3}{(n+1)^3} \right) = \frac{n+1}{3} \sum_{k=0}^n p_{n,k}(x) \frac{3k^2+2k+1}{(n+1)^3} = \\
&= \frac{1}{3(n+1)^2} \sum_{k=0}^n p_{n,k}(x) (3k^2+2k+1) = \frac{n^2}{(n+1)^2} \sum_{k=0}^n p_{n,k}(x) \frac{k^2}{n^2} + \frac{n}{(n+1)^2} \sum_{k=0}^n p_{n,k}(x) \frac{k}{n} + \\
&+ \frac{1}{3(n+1)^2} \sum_{k=0}^n p_{n,k}(x) = \frac{n^2}{(n+1)^2} B_n(e_2)(x) + \frac{n}{(n+1)^2} B_n(e_1)(x) + \frac{1}{3(n+1)^2} B_n(e_0)(x) = \\
&= \frac{n^2}{(n+1)^2} \left(x^2 + \frac{x(1-x)}{n} \right) + \frac{n}{(n+1)^2} x + \frac{1}{3(n+1)^2}
\end{aligned}$$

(c) Observăm că pentru orice $f \in C[0, 1]$ și $x \in [0, 1]$ avem

$$U_n(f)(x) = \frac{1}{3} B_n(f)(x) + \frac{2}{3} K_n(f)(x)$$

Din liniaritatea și pozitivitatea lui B_n și K_n deducem că U_n este un operator liniar și pozitiv. În continuare vom calcula $U_n(e_0)$, $U_n(e_1)$, $U_n(e_2)$.

$$\begin{aligned}
U_n(e_0) &= \frac{1}{3} B_n(e_0) + \frac{2}{3} K_n(e_0) = \frac{1}{3} e_0 + \frac{2}{3} e_0 = e_0 \\
\\
U_n(e_1) &= \frac{1}{3} B_n(e_1) + \frac{2}{3} K_n(e_1) = \frac{1}{3} B_n(e_1) + \frac{2}{3} \left(\frac{n}{n+1} B_n(e_1) + \frac{1}{2(n+1)} B_n(e_0) \right) = \\
&= \frac{3n+1}{3(n+1)} B_n(e_1) + \frac{1}{3(n+1)} B_n(e_0) = \frac{3n+1}{3(n+1)} e_1 + \frac{1}{3(n+1)} e_0 \\
\\
U_n(e_2) &= \frac{1}{3} B_n(e_2) + \frac{2}{3} K_n(e_2) = \frac{1}{3} B_n(e_2) + \frac{2}{3} \left(\frac{n^2}{(n+1)^2} B_n(e_2) + \frac{n}{(n+1)^2} B_n(e_1) + \frac{1}{3(n+1)^2} B_n(e_0) \right) = \\
&= \frac{3n^2+2n+1}{3(n+1)^2} B_n(e_2) + \frac{2n}{3(n+1)^2} B_n(e_1) + \frac{2}{9(n+1)^2} B_n(e_0) = \frac{3n^2+2n+1}{3(n+1)^2} \frac{(n-1)e_2 + e_1}{n} + \\
&+ \frac{2n}{3(n+1)^2} e_1 + \frac{2}{9(n+1)^2} e_0 = \frac{3n^3-n^2-n-1}{3n(n+1)} e_2 + \frac{5n^2+2n+1}{3n(n+1)^2} e_1 + \frac{2}{9(n+1)^2} e_0
\end{aligned}$$

2. Fie $p_{n,k}(x)$ polinoamele fundamentale Bernstein. Să se demonstreze următoarea relație de recurență

$$p_{n,k}(x) = (1-x)p_{n-1,k}(x) + xp_{n-1,k-1}(x)$$

Rezolvare: Știm că $p_{n,k}(x) = C_n^k x^k (1-x)^{n-k}$. Avem

$$\begin{aligned} (1-x)p_{n-1,k}(x) + xp_{n-1,k-1}(x) &= \\ &= (1-x)C_{n-1}^k x^k (1-x)^{n-1-k} + xC_{n-1}^{k-1} x^{k-1} (1-x)^{n-k} = \\ &= C_{n-1}^k x^k (1-x)^{n-k} + C_{n-1}^{k-1} x^k (1-x)^{n-k} = x^k (1-x)^{n-k} (C_{n-1}^k + C_{n-1}^{k-1}) = \\ &= C_n^k x^k (1-x)^{n-k} = p_{n,k}(x) \end{aligned}$$

3. Să se calculeze $B_n(e_3)(x)$, unde $B_n : C[0,1] \rightarrow C[0,1]$ sunt operatorii Bernstein, iar $e_k(x) = x^k$

Rezolvare:

$$B_n(f)(x) = \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} f\left(\frac{k}{n}\right)$$

Pentru $e_0(x) = 1$ avem

$$B_n(e_0)(x) = \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} = (x + (1-x))^n = 1$$

Considerăm identitatea

$$(p+q)^n = \sum_{k=0}^n C_n^k p^k q^{n-k}$$

și derivând în funcție de p obținem

$$\frac{d}{dp}((p+q)^n) = n(p+q)^{n-1} = \sum_{k=0}^n C_n^k k p^{k-1} q^{n-k}.$$

Pasul următor constă în a înmulți cu $\frac{p}{n}$ și deducem că

$$\sum_{k=0}^n C_n^k p^k q^{n-k} \left(\frac{k}{n}\right) = p(p+q)^{n-1}.$$

Înlocuind $p = x$ și $q = 1-x$ ajungem la

$$B_n(e_1)(x) = \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} \left(\frac{k}{n}\right) = x(x+1-x)^{n-1} = x$$

Derivând expresia de mai sus de două ori în raport cu p și înmulțind de fiecare dată cu $\frac{p}{n}$ obținem

$$\sum_{k=0}^n C_n^k p^k q^{n-k} \left(\frac{k}{n}\right)^2 = \frac{(n-1)(p+q)^{n-2}}{n} p^2 + \frac{(p+q)^{n-1}}{n} p$$

și

$$\sum_{k=0}^n C_n^k p^k q^{n-k} \left(\frac{k}{n}\right)^3 = \frac{(n-1)(n-2)(p+q)^{n-3}}{n^2} p^3 + \frac{3(n-1)(p+q)^{n-2}}{n^2} p^2 + \frac{(p+q)^{n-1}}{n^2} p$$

Luând $p = x$ și $q = 1-x$ deducem că

$$B_n(e_2)(x) = \frac{(n-1)x^2}{n} + \frac{x}{n}$$

și

$$B_n(e_3)(x) = \frac{(n-1)(n-2)x^3}{n^2} + \frac{3(n-1)x^2}{n^2} + \frac{x}{n^2}$$

4. Să se calculeze $B_n(f)(x)$ unde $f(x) = \cos(x)$

Rezolvare: Avem

$$B_n(\cos)(x) = \sum_{k=0}^n p_{n,k}(x) \cos\left(\frac{k}{n}\right) = \sum_{k=0}^n x^k (1-x)^{n-k} \cos\left(\frac{k}{n}\right)$$

Considerăm sumele

$$S_1 = \sum_{k=0}^n x^k (1-x)^{n-k} \cos \frac{k}{n}$$

$$S_2 = \sum_{k=0}^n x^k (1-x)^{n-k} \sin \frac{k}{n}$$

Deducem că $B_n(\cos)(x)$ va fi partea reală a sumei $S_1 + iS_2$

$$\begin{aligned} S_1 + iS_2 &= \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} \left(\cos \frac{k}{n} + i \sin \frac{k}{n} \right) = \\ &= \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} \left(\cos \frac{1}{n} + i \sin \frac{1}{n} \right)^k = \\ &= \sum_{k=0}^n C_n^k \left(x \left(\cos \frac{1}{n} + i \sin \frac{1}{n} \right) \right)^k (1-x)^{n-k} = \\ &= \left(x \left(\cos \frac{1}{n} + i \sin \frac{1}{n} \right) + 1 - x \right)^n = \left(x \cos \frac{1}{n} + 1 - x + ix \sin \frac{1}{n} \right)^n \end{aligned}$$

adică

$$\begin{aligned} B_n(\cos)(x) &= C_n^0 \left(x \cos \frac{1}{n} + 1 - x \right)^n + (-1) C_n^2 \left(x \cos \frac{1}{n} + 1 - x \right)^{n-2} \left(x \sin \frac{1}{n} \right)^2 \\ &+ (-1)^2 C_n^4 \left(x \cos \frac{1}{n} + 1 - x \right)^{n-4} \left(x \sin \frac{1}{n} \right)^4 + \dots = \\ &= \sum_{k=0}^n (-1)^k C_n^{2k} \left(x \cos \frac{1}{n} + 1 - x \right)^{n-2k} \left(x \sin \frac{1}{n} \right)^{2k} \end{aligned}$$

Rezolvare II: Folosim formula $\cos x = \frac{e^{ix} + e^{-ix}}{2}$ și liniaritatea operatorului Bernstein B_n și obținem

$$\begin{aligned} B_n(\cos x)(x) &= \frac{1}{2} B_n(e^{ix})(x) + \frac{1}{2} B_n(e^{-ix})(x) = \frac{1}{2} \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} e^{i \frac{k}{n}} + \\ &+ \frac{1}{2} \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} e^{-i \frac{k}{n}} = \frac{1}{2} \sum_{k=0}^n C_n^k \left(x e^{i \frac{1}{n}} \right)^k (1-x)^{n-k} + \frac{1}{2} \sum_{k=0}^n C_n^k \left(x e^{-i \frac{1}{n}} \right)^k (1-x)^{n-k} = \\ &= \frac{1}{2} \left(x e^{i \frac{1}{n}} + 1 - x \right)^n + \frac{1}{2} \left(x e^{-i \frac{1}{n}} + 1 - x \right)^n = \operatorname{Re} \left(x e^{i \frac{1}{n}} + 1 - x \right)^n \end{aligned}$$